

There are a total of six problems. You have to solve all the problems.

Problem 1 (CO1): Finite Automata (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{000, 010, 100, 110\}$$

$$L_2 = \{w : w \text{ contains exactly two 1s between every successive 0s}\}$$

$$L_3 = L_1^* \cap L_2$$

$$L_4 = \{w : w \text{ contains } x \text{ as a substring where } x \in L_3 \text{ and } |x| > 0\}$$

Now solve the following problems.

- Give** the state diagram of a DFA that recognizes L_1^* . [Note: L_1 is a finite language consisting of four strings] (3 points)
- Give** the state diagram of a DFA that recognizes L_2 . (3 points)
- Write** all strings of length at most ten in L_3 . (1 point)
- What** is the minimum number of states required to construct an NFA for L_3 . (1 point)
- Give** the state diagram of a DFA that recognizes L_4 . [Recall: $|x|$ denotes the length of the string x] (2 points)

Problem 2 (CO2): Converting Regular Expressions to NFA (10 points)

Convert the following regular expression over $\Sigma = \{a, b\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

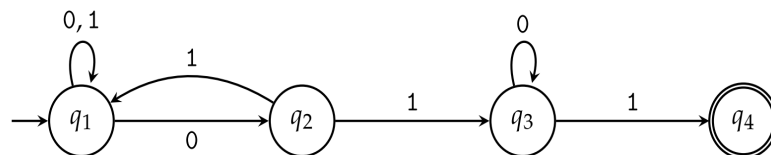
$$bb(b + (a + bb)^*)^*a$$

Problem 3 (CO1): Regular Languages (5 points)

Answer the following questions.

- For $\Sigma = \{0, 1, 2\}$, **what** is the total number of transitions in a DFA with 8 states? (1 point)
- Write** the regular expression for $R.\phi$, where R is a regular expression and ϕ denotes the empty language. (1 point)
- Write** the smallest string generated by the following regular expression: $aa(aaa + ba)(a + \epsilon) + ba^*a(ab + (aa)^*b)$. (1 point)
- Write** whether the regular expressions $1^*(011^*)^*(0 + \epsilon)$ and $(0 + \epsilon)(10 + 1)^*$ are equivalent or not. (1 point)

Consider the following NFA:



Now consider the following the regular languages:

$$L_1 = \{w \in \{0, 1\}^* : w \text{ contains at least two 1s}\}$$

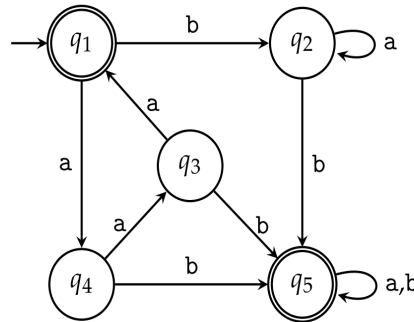
$$L_2 = \{w \in \{0, 1\}^* : w \text{ contains the substring } 011\}$$

$$L_3 = \{w \in \{0, 1\}^* : w \text{ ends with } 1\}$$

- Write** the correct option below that specifies the languages recognized by the given NFA. (1 point)
- (i) Only L_2 (ii) Only L_3 (iii) L_2 and L_3 (iv) L_1, L_2 and L_3 (v) None of L_1, L_2 and L_3

Problem 4 (CO2): Converting FA to Regular Expressions (10 points)

Convert the following finite automata into an equivalent regular expression using the state elimination method. You must eliminate the states in the following order: q_2, q_3, q_4, q_1, q_5 . Show each step of the process.



Problem 5 (CO1): Regular Expressions (10 points)

Let $\Sigma = \{0, 1\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ starts and ends with } 1\}$$

$$L_2 = \{w : |w| \% 3 = 0\}$$

$$L_3 = \{w = 0^n 1^m : n \text{ is even and } m \text{ is odd}\}$$

$$L_4 = \{w : w \text{ contains even number of } 01 \text{ as a substring}\}$$

$$L_5 = L_1 \cap L_2$$

Now solve the following problems.

- Give** a regular expression that generates L_1 . (2 points)
- Give** a regular expression that generates $\overline{L_2}$. [Recall: $\overline{L}$ denotes the complement of the language L i.e., $\overline{L} = \Sigma^* - L$] (2 points)
- Give** a regular expression that generates L_3 . (2 points)
- Give** a regular expression that generates L_4 . (2 points)
- Give** a regular expression that generates L_5 . (2 points)

Problem 6 (CO2): Converting NFA to DFA (5 points)

Write whether the following statements are true or false. In addition, write the correct statements for the false statements.

- When converting an NFA to an equivalent DFA using the subset construction method, the resulting DFA can be exponentially larger than the original NFA. (1 point)
- Using the subset construction method it can be proved that every language that is recognized by a DFA can also be recognized by a NFA. (1 point)

Consider the following NFA:



Now answer the following questions.

- Write** the ϵ -closure of state q_1 in the given NFA. (1 point)
- Write** the subset of NFA states that will form the starting state of the equivalent DFA. (1 point)
- What** is $\delta(\{q_2, q_4\}, 1)$ in the given NFA? [Recall: $\delta(\{q\}, a)$ outputs the set of states in which the NFA transitions when it is in state q and reads the symbol a] (1 point)